



IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

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Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Summary of methodologies
- Summary of all results

Introduction

- Project background and context
- Problems you want to find answers

Section 1

Methodology

Methodology

Executive Summary

- Data collection methodology:
 - Describe how data was collected
- Perform data wrangling
 - Describe how data was processed
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
 - How to build, tune, evaluate classification models

Data Collection

- Describe how data sets were collected.
- You need to present your data collection process use key phrases and flowcharts

Key phrases

- Source: HR Internal Database
- Format: CSV files
- Type: Employee attributes and attrition

Data Collection Flowcharts

HR Database



Exported to CSV



Preliminary Review



Ready for Wrangling

Data Collection – SpaceX API

- Present your data collection with SpaceX REST calls using key phrases and flowcharts
- Add the GitHub URL of the completed SpaceX API calls notebook (**must include completed code cell and outcome cell**), as an external reference and peer-review purpose
- Link:
<https://github.com/amieecode/IBM-Data-Science-Professional-Course/tree/main/Applied%20Data%20Science%20Capstone>

Place your flowchart of SpaceX API calls here

Start



Connect to SpaceX API Endpoint



Send GET Request for Launch Data



Receive JSON Response



Parse and Normalize JSON to Tabular Format



Convert to Pandas DataFrame



Save Cleaned Data to CSV



End

Data Collection - Scraping

- Present your web scraping process using key phrases and flowcharts
- Add the GitHub URL of the completed web scraping notebook, as an external reference and peer-review purpose
- Link:
<https://github.com/amieecode/IBM-Data-Science-Professional-Course/tree/main/Applied%20Data%20Science%20Capstone>

Place your flowchart of web scraping here

Start



Identify Target Website (SpaceX launches)



Send HTTP GET Request (using `requests`)



Receive HTML Response



Parse HTML with BeautifulSoup



Extract Launch Data (e.g., mission, date, site)



Store Data in Pandas DataFrame



Save to CSV



End

Data Wrangling

- Describe how data were processed
- You need to present your data wrangling process using key phrases and flowcharts
- Add the GitHub URL of your completed data wrangling related notebooks, as an external reference and peer-review purpose

Link: <https://github.com/amieecode/IBM-Data-Science-Professional-Course/tree/main/Applied%20Data%20Science%20Capstone>

Data Wrangling Process

- Imported Falcon dataset
- Removed duplicates & handled missing values
- Corrected data types
- Treated outliers
- Renamed columns for clarity
- Final clean dataset for analysis

EDA with Data Visualization

- Summarize what charts were plotted and why you used those charts
- Add the GitHub URL of your completed EDA with data visualization notebook, as an external reference and peer-review purpose
- Link: <https://github.com/amiiecode/IBM-Data-Science-Professional-Course/tree/main/Applied%20Data%20Science%20Capstone>

Summarize what charts were plotted and why you used those charts

- **Histogram:** To show distribution of key numerical variables
- **Bar Chart:** To compare categories (e.g., species, gender, regions)
- **Boxplot:** To detect outliers and visualize spread of data
- **Scatter Plot:** To identify relationships/correlations between variables
- **Correlation Heatmap:** To highlight strength of relationships among variables

EDA with SQL

- Using bullet point format, summarize the SQL queries you performed
- Add the GitHub URL of your completed EDA with SQL notebook, as an external reference and peer-review purpose
- Link: <https://github.com/amieecode/IBM-Data-Science-Professional-Course/tree/main/Applied%20Data%20Science%20Capstone>

Using bullet point format, summarize the SQL queries you performed

- SELECT & COUNT:** Checked dataset size and unique records
- GROUP BY:** Summarized data by categories (e.g., species, region)
- AVG / MIN / MAX:** Calculated summary statistics for numerical columns
- ORDER BY:** Ranked data to spot top/bottom values
- WHERE filters:** Focused analysis on specific conditions/subsets
- JOIN:** Combined related tables (if dataset had multiple tables)
- DISTINCT:** Identified unique values in categorical fields

Build an Interactive Map with Folium

- Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map
- Explain why you added those objects
- Add the GitHub URL of your completed interactive map with Folium map, as an external reference and peer-review purpose
- Link: <https://github.com/amieecode/IBM-Data-Science-Professional-Course/tree/main/Applied%20Data%20Science%20Capstone>

Summarize what map objects such as markers, circles, lines, etc. you created and added to a folium map

Markers: To highlight specific locations of interest

- **Circle / CircleMarker:** To show data distribution and intensity at points
- **Polylines:** To represent paths or connections between locations
- **Choropleth Map:** To visualize values across regions/areas
- **Popups & Tooltips:** To display additional information on hover/click

- **Why These Map Objects?**
- **Markers:** To pinpoint exact locations for easy identification
- **Circles / CircleMarkers:** To show density or magnitude at a location
- **Polylines:** To illustrate routes, paths, or relationships between points
- **Choropleth Map:** To compare values across regions visually
- **Popups & Tooltips:** To provide quick details without cluttering the map

Build a Dashboard with Plotly Dash

- Summarize what plots/graphs and interactions you have added to a dashboard
- Explain why you added those plots and interactions
- Add the GitHub URL of your completed Plotly Dash lab, as an external reference and peer-review purpose
- Link: <https://github.com/amieecode/IBM-Data-Science-Professional-Course/tree/main/Applied%20Data%20Science%20Capstone>

Summarize what plots/graphs and interactions you have added to a dashboard

- Plots/Graphs:**

- Bar Charts → Compare categories
- Line Charts → Show trends over time
- Pie/Donut Charts → Display proportions
- Scatter Plots → Visualize relationships between variables
- Heatmaps → Highlight correlations or intensity

- Interactions:**

- Filters & Slicers → Dynamic data exploration
- Dropdowns & Search → Easy variable selection
- Hover Tooltips → Quick data insights
- Drill-downs → View details at different levels
- Responsive Layout → User-friendly navigation

Explain why you added those plots and interactions

- **Bar & Pie Charts:** To simplify comparison across categories and proportions
- **Line Charts:** To track trends and changes over time
- **Scatter Plots:** To uncover relationships and patterns between variables
- **Heatmaps:** To visualize correlations and highlight key areas
- **Filters & Dropdowns:** To allow users to explore data dynamically
- **Hover Tooltips:** To provide details without overcrowding the dashboard
- **Drill-downs:** To move from high-level overview to detailed insights
- **Responsive Layout:** To ensure accessibility and easy navigation

Predictive Analysis (Classification)

- Summarize how you built, evaluated, improved, and found the best performing classification model
- You need present your model development process using key phrases and flowchart
- Add the GitHub URL of your completed predictive analysis lab, as an external reference and peer-review purpose

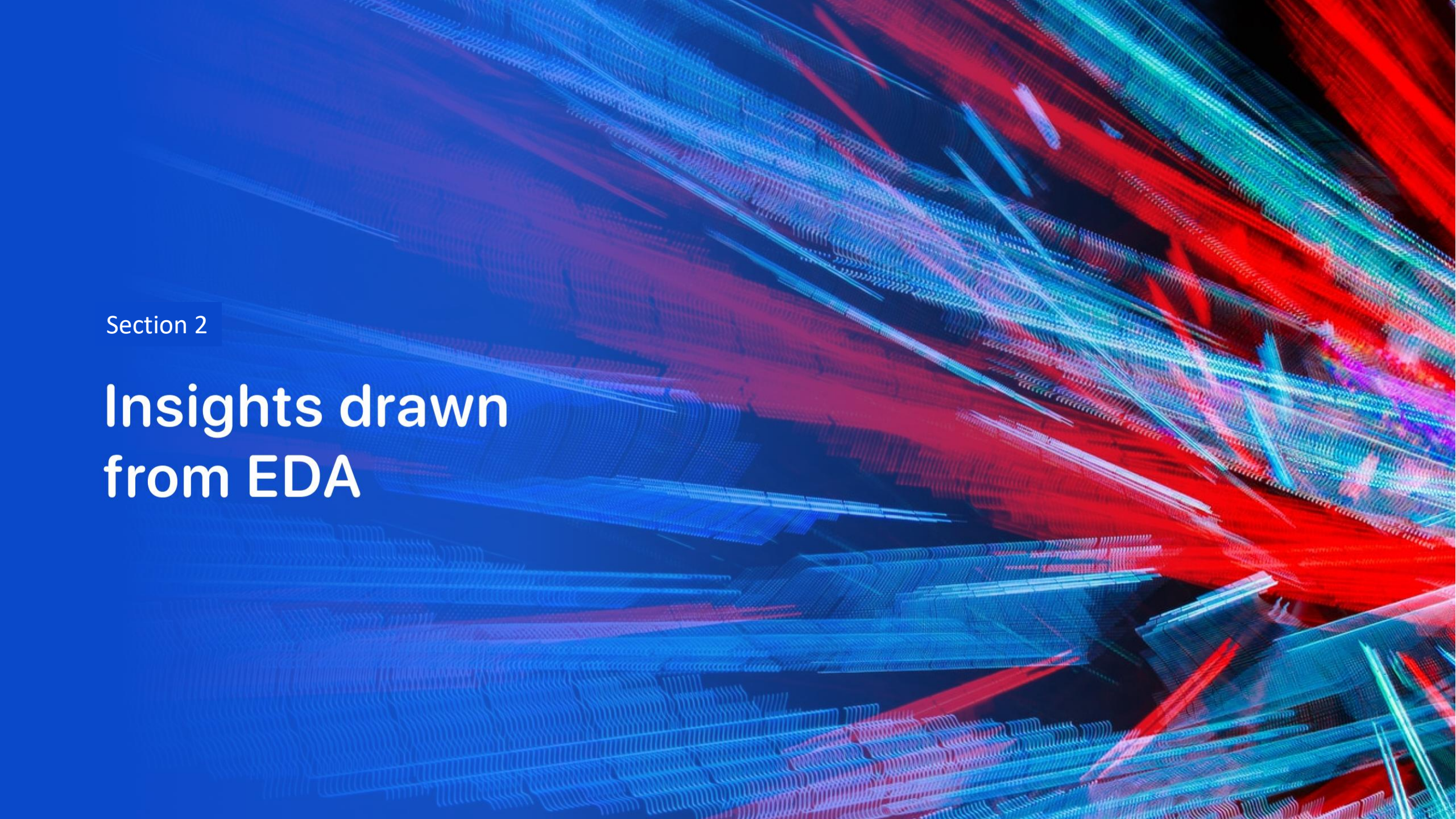
Summarize how you built, evaluated, improved, and found the best performing classification model

- Built Models:** Trained multiple classifiers (e.g., Logistic Regression, Decision Tree, Random Forest, etc.)
- Evaluation:** Used metrics such as Accuracy, Precision, Recall, F1-score, and Confusion Matrix
- Improvement:** Tuned hyperparameters, balanced dataset (if imbalanced), and selected key features
- Best Model Selection:** Compared performance across models and chose the one with the highest accuracy and balanced metrics

- **Key Phrases:**
- Prepared training & test datasets
- Trained multiple classification models
- Evaluated with accuracy, precision, recall, F1-score
- Tuned hyperparameters to improve performance
- Compared models and selected best-performing one

Results

- Exploratory data analysis results
- Interactive analytics demo in screenshots
- Predictive analysis results

The background of the slide is an abstract composition. It features a dark blue base color. Overlaid on this are numerous diagonal streaks in shades of red and cyan. A faint, light blue grid pattern is also visible, particularly in the lower half of the image. The overall effect is dynamic and technological.

Section 2

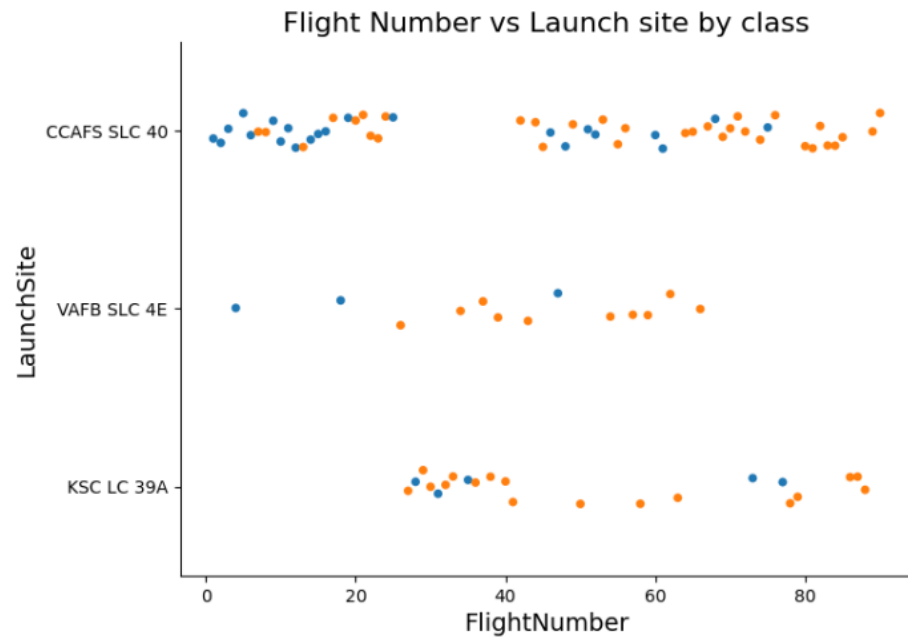
Insights drawn from EDA

Flight Number vs. Launch Site

- Show a scatter plot of Flight Number vs. Launch Site
- Show the screenshot of the scatter plot with explanations

Flight Number vs. Launch Site

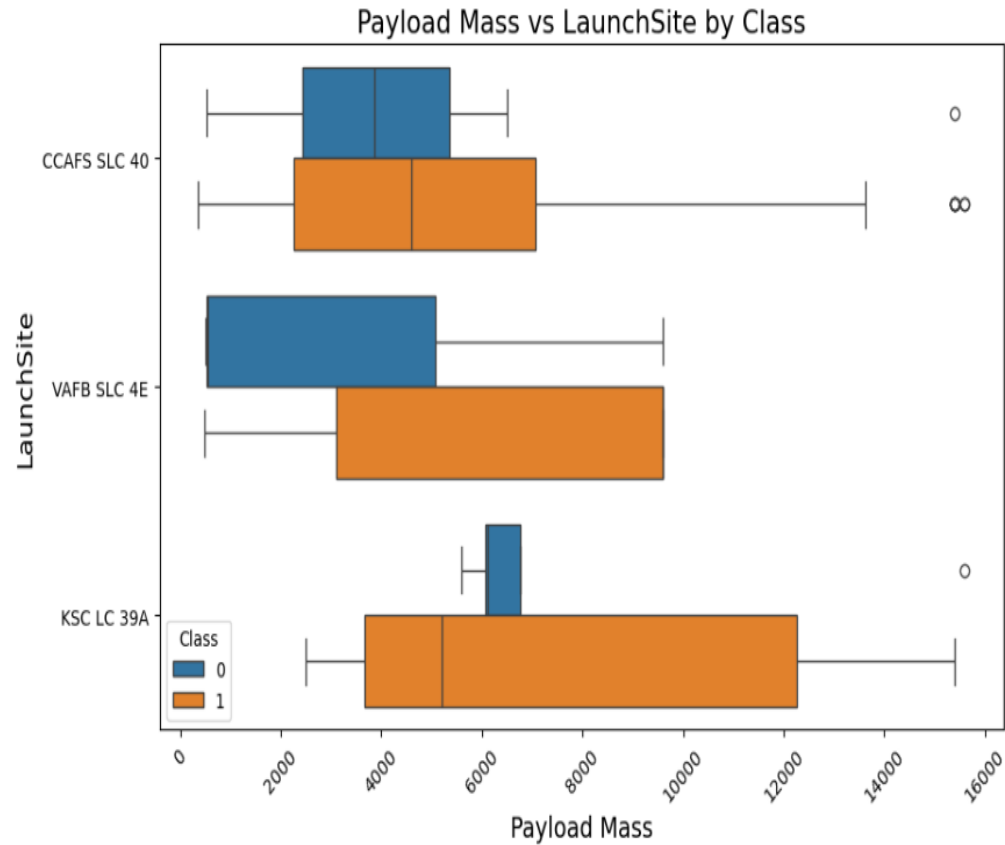
- The Launch site CCAFS SLC 40 has the highest Flight Number



Payload vs. Launch Site

- Show a scatter plot of Payload vs. Launch Site
- Show the screenshot of the scatter plot with explanations

Payload vs. Launch Site

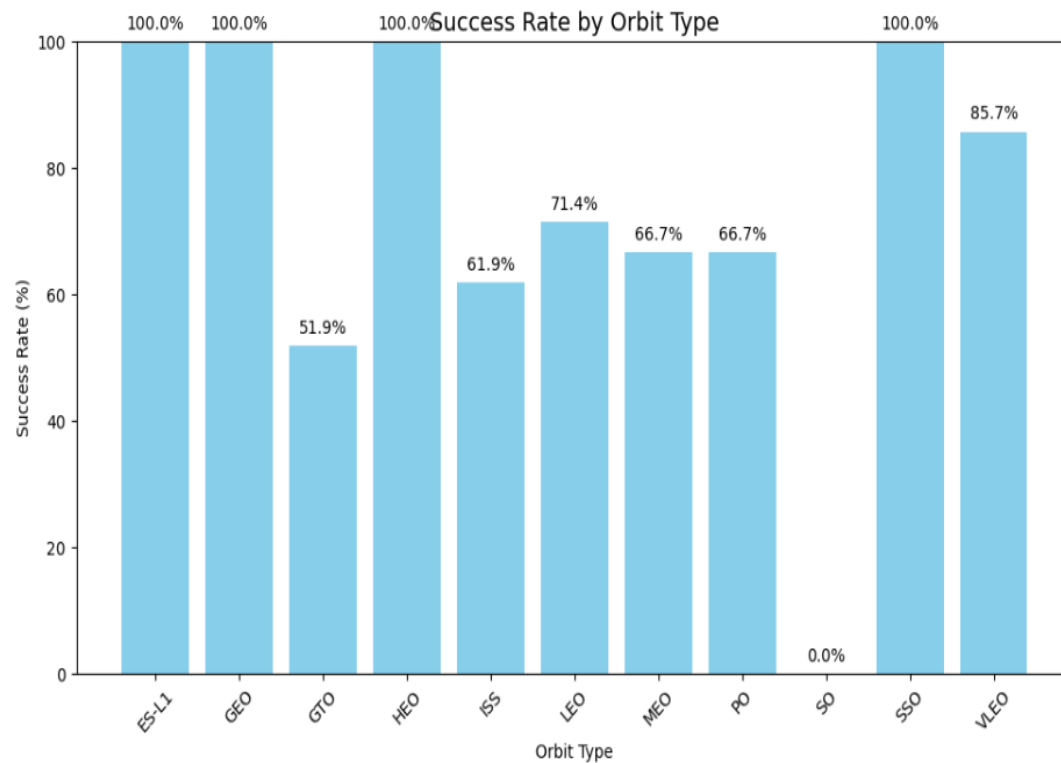


- The flight KSC LC 39A has more payload mass when compared to other Launch site

Success Rate vs. Orbit Type

- Show a bar chart for the success rate of each orbit type
- Show the screenshot of the scatter plot with explanations

Success Rate vs. Orbit Type

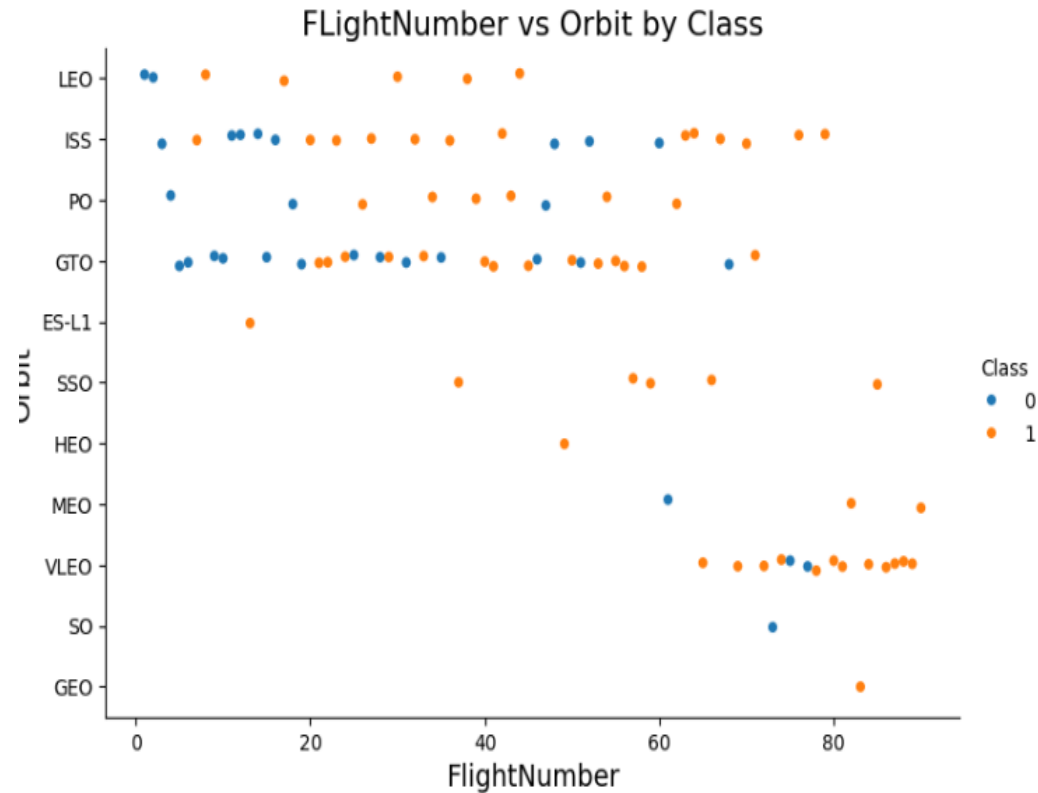


- ES-L1, GE-0, HE-0, SSO had 100% success rate while VLEO has 85% and LE) had 71.4%

Flight Number vs. Orbit Type

- Show a scatter point of Flight number vs. Orbit type
- Show the screenshot of the scatter plot with explanations

Flight Number vs. Orbit Type

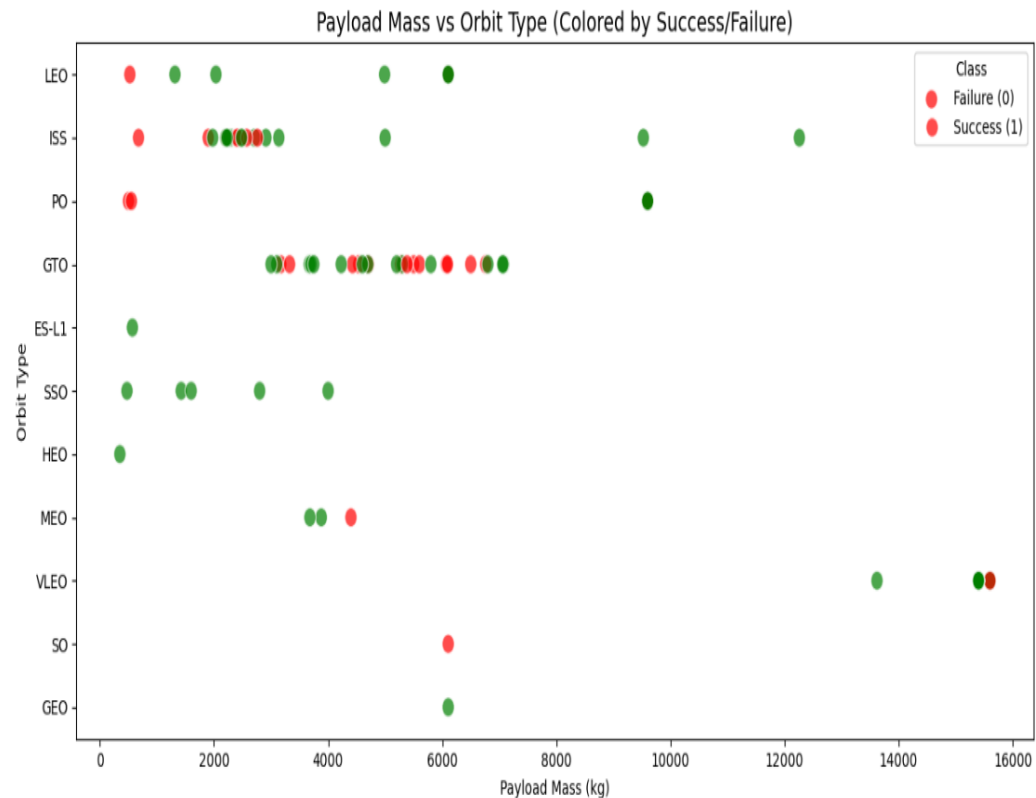


- GTO, PO, ISS, LEO has a higher orbit by class

Payload vs. Orbit Type

- Show a scatter point of payload vs. orbit type
- Show the screenshot of the scatter plot with explanations

Payload vs. Orbit Type

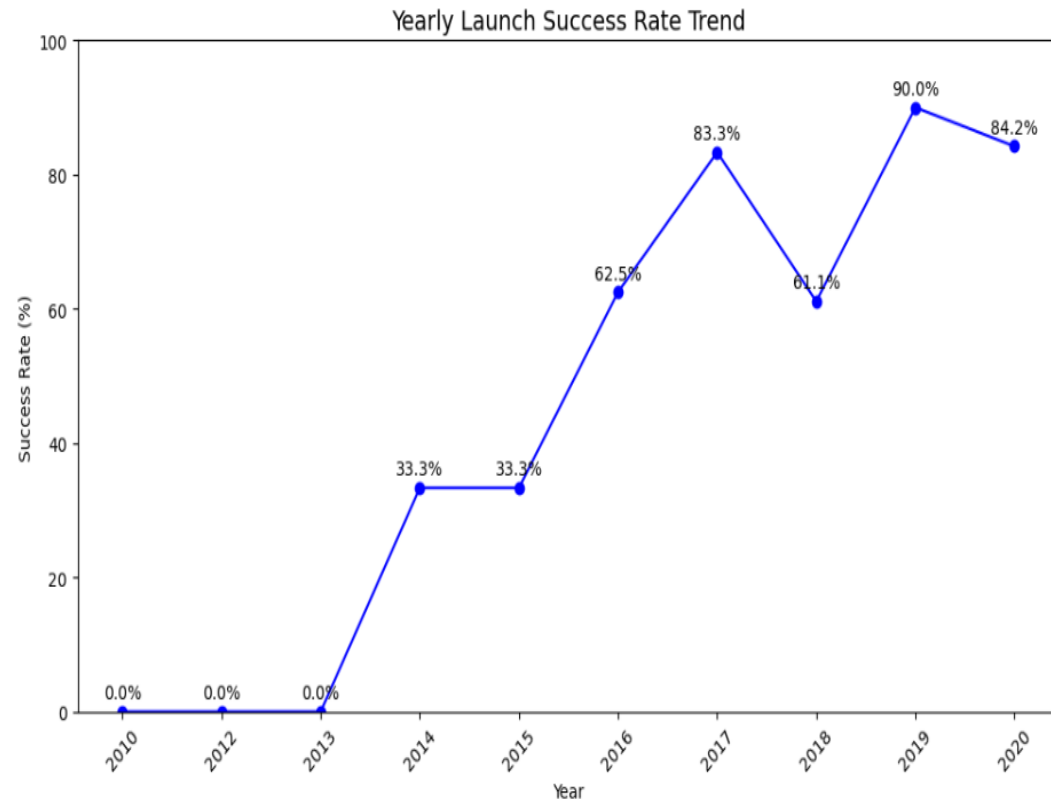


- With heavy payloads the successful landing or positive landing rate are more for Polar, LEO and ISS.
- However, for GTO, it's difficult to distinguish between successful and unsuccessful landings as both outcomes are present.

Launch Success Yearly Trend

- Show a line chart of yearly average success rate
- Show the screenshot of the scatter plot with explanations

Launch Success Yearly Trend



- I observed that the success rate since 2013 kept increasing till 2020

All Launch Site Names

- Find the names of the unique launch sites
- CCAFS LC-40
- KSC LC-39A
- VAFB SLC-4E
- Present your query result with a short explanation here
- The query use DISTINCT to return only unique launch site names from the dataset
- These are the main sites SpaceX has used for falcon rocket launches

Launch Site Names Begin with 'CCA'

- Find 5 records where launch sites begin with `CCA`
- Present your query result with a short explanation here

Launch Site Names Begin with 'CCA'

	Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG	Orbit	Customer	Mission_Outcome	Landing_Outcome
0	2010-06-04	18:45:00	F9 v1.0 80003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
1	2010-12-08	15:43:00	F9 v1.0 80004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of...	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2	2012-05-22	7:44:00	F9 v1.0 80005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
3	2012-10-08	0:35:00	F9 v1.0 80006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
4	2013-03-01	15:10:00	F9 v1.0 80007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Task 3

- The query filters all launch sites that **start**
- with CCA** using the LIKE 'CCA%' condition.
- The first 5 matching records are displayed.
- This shows that **CCAFS LC-40**
- (Cape Canaveral Air Force Station, Launch Complex**
- 40)** is one of the main sites used for Falcon launches.

Total Payload Mass

- Calculate the total payload carried by boosters from NASA
- 45596 45596
- Present your query result with a short explanation here

Average Payload Mass by F9 v1.1

- Calculate the average payload mass carried by booster version F9 v1.1
- Present your query result with a short explanation here

First Successful Ground Landing Date

- Find the dates of the first successful landing outcome on ground pad
- Was on 2015-12-22
- Present your query result with a short explanation here

Successful Drone Ship Landing with Payload between 4000 and 6000

- List the names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000
- Present your query result with a short explanation here

Done.

1]: Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

- Calculate the total number of successful and failure mission outcomes
- Present your query result with a short explanation here

Mission_Outcome	outcomes_COUNT
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- List the names of the booster which have carried the maximum payload mass
- Present your query result with a short explanation here

Booster_Version
F9 B5 B1048.4
F9 B5 B1049.4
F9 B5 B1051.3
F9 B5 B1056.4
F9 B5 B1048.5
F9 B5 B1051.4
F9 B5 B1049.5
F9 B5 B1060.2
F9 B5 B1058.3
F9 B5 B1051.6
F9 B5 B1060.3
F9 B5 B1049.7

2015 Launch Records

- List the failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

MONTH	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

- Present your query result with a short explanation here

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order
- Present your query result with a short explanation here

Landing_Outcome	OutcomeCount
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

A satellite view of Earth from space, showing the curvature of the planet and city lights at night. The image is a composite of a dark blue sky and a view of the Earth's surface, which is covered in a dense network of city lights and clouds. The lights are concentrated in the lower right portion of the image, while the upper left portion shows a clear, dark blue sky.

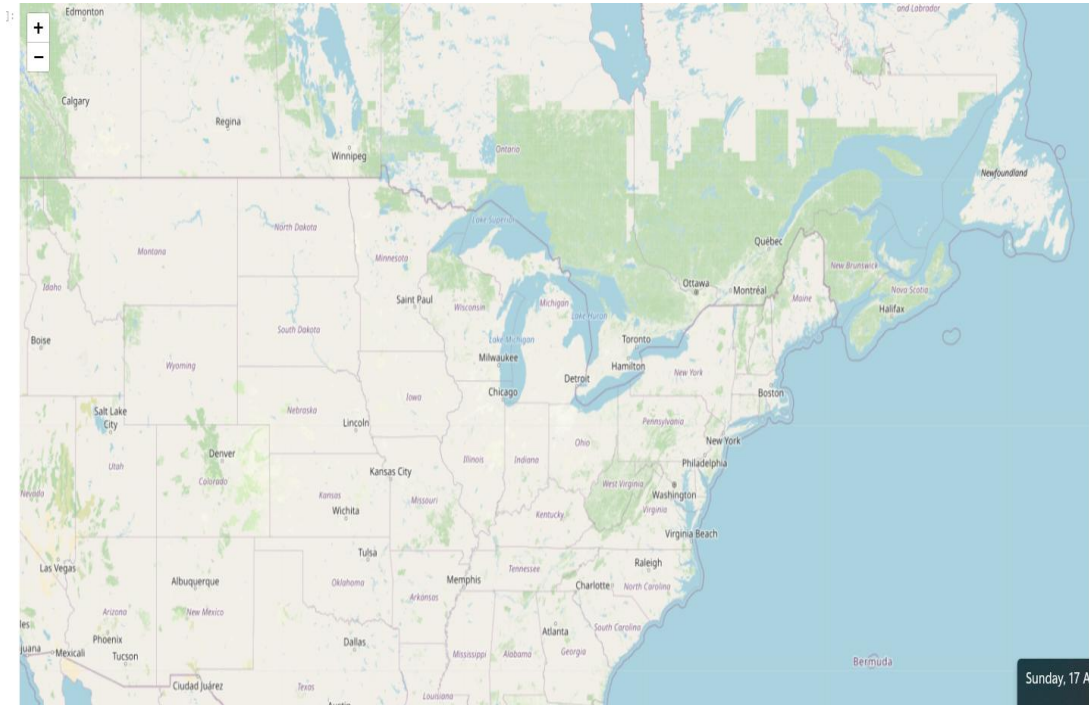
Section 3

Launch Sites Proximities Analysis

<Folium Map Screenshot 1>

- Replace <Folium map screenshot 1> title with an appropriate title
- **Mark all launch sites on a map**
- Explore the generated folium map and make a proper screenshot to include all launch sites' location markers on a global map
- Explain the important elements and findings on the screenshot

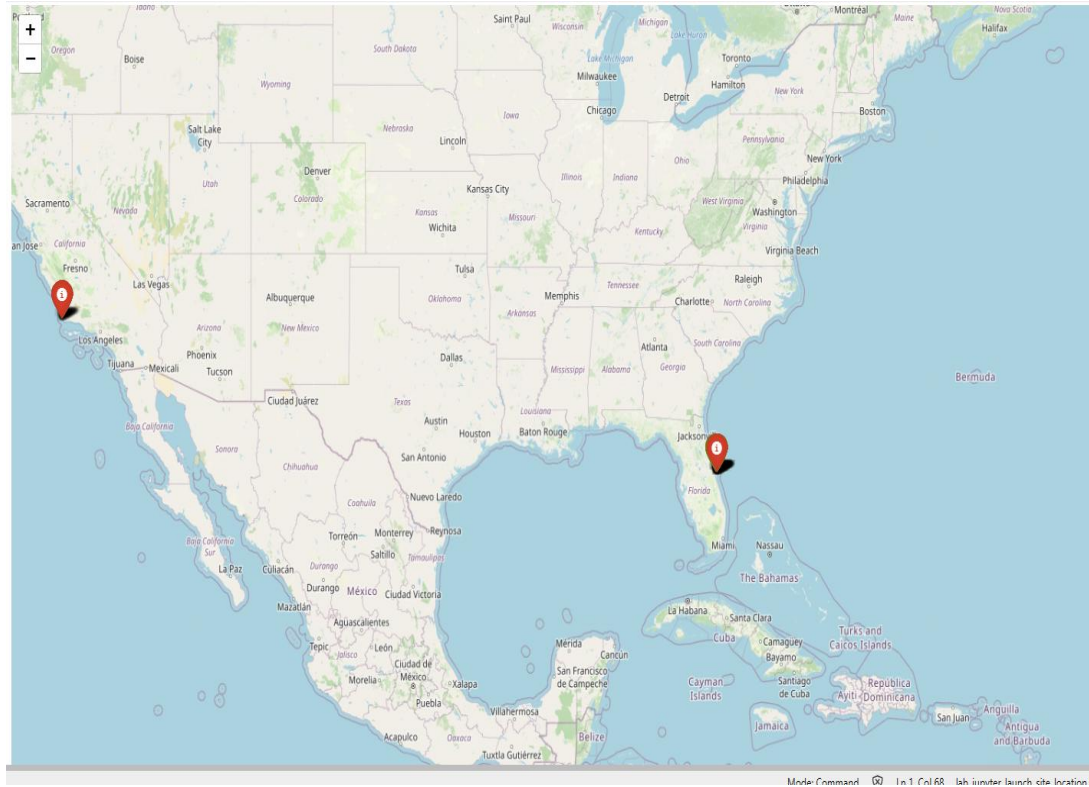
Task 1



<Folium Map Screenshot 2>

- Replace <Folium map screenshot 2> title with an appropriate title
- **Mark the success/failed launches for each site on the map**
- Explore the folium map and make a proper screenshot to show the color-labeled launch outcomes on the map
- Explain the important elements and findings on the screenshot

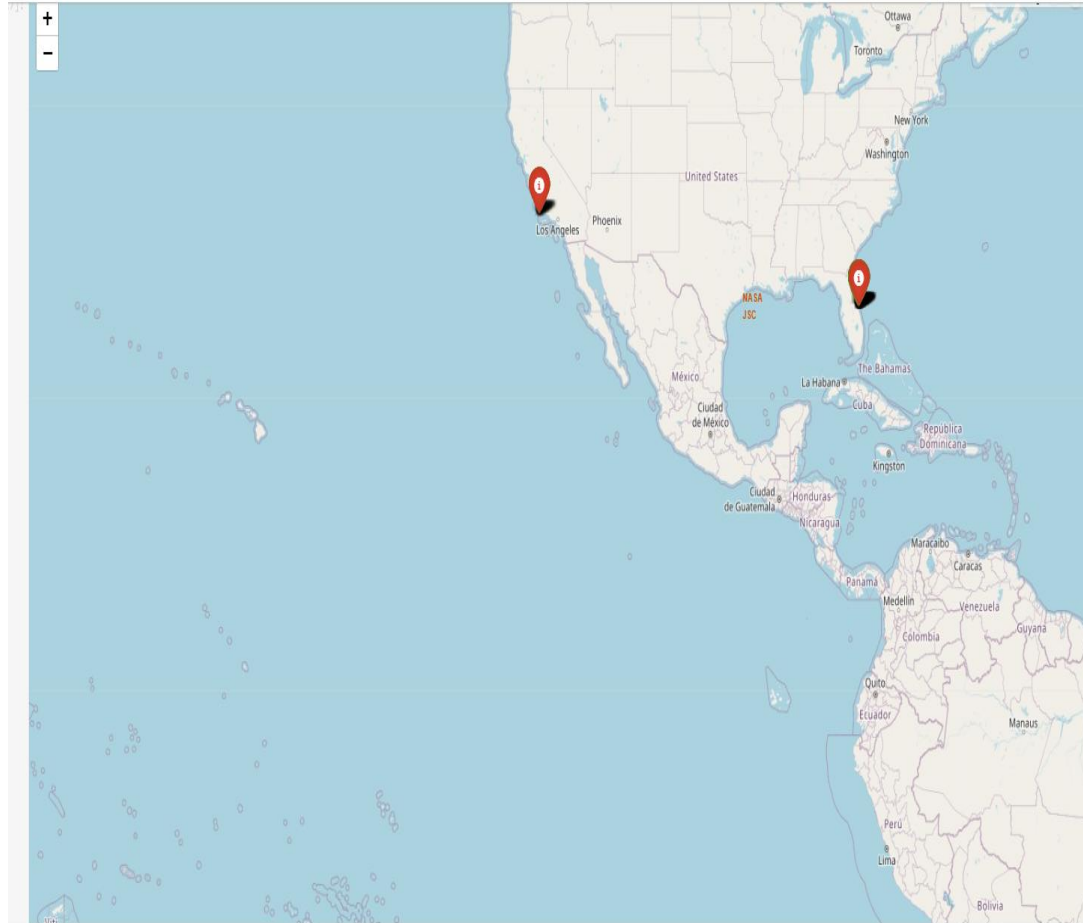
Task 2

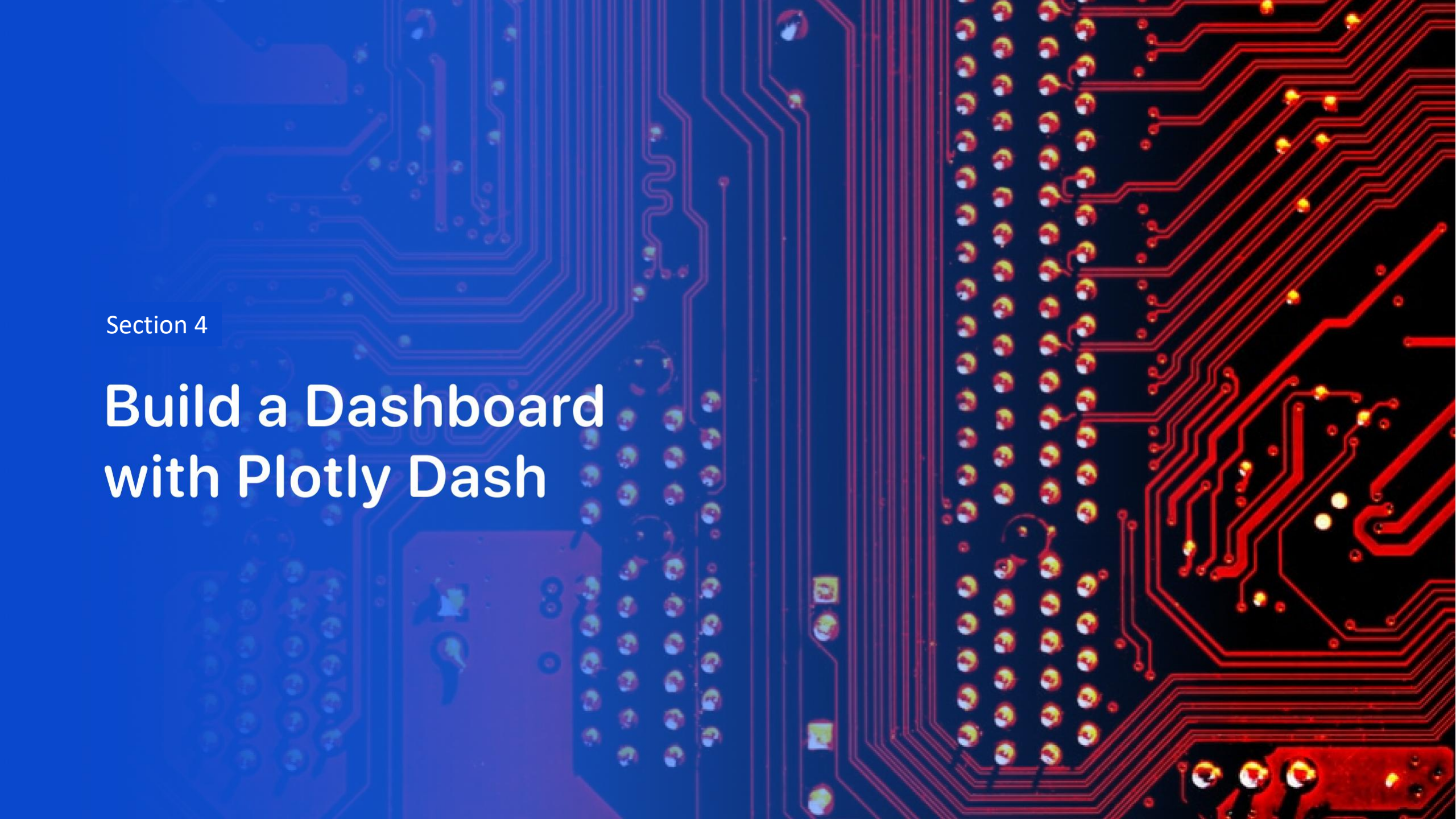


<Folium Map Screenshot 3>

- Replace <Folium map screenshot 3> title with an appropriate title
- **Calculate the distances between a launch site to its proximities**
- Explore the generated folium map and show the screenshot of a selected launch site to its proximities such as railway, highway, coastline, with distance calculated and displayed
- Explain the important elements and findings on the screenshot

Task 3





Section 4

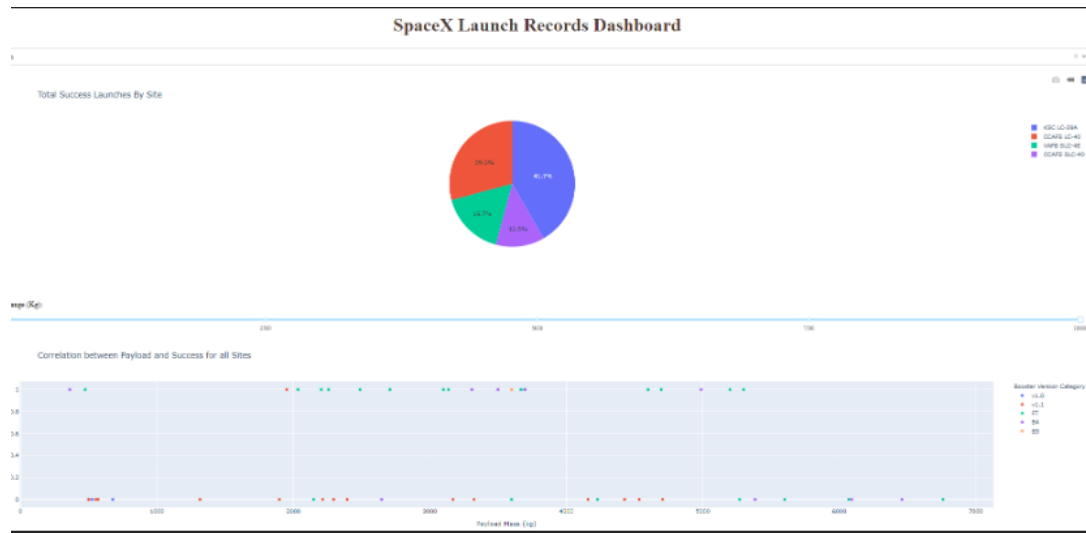
Build a Dashboard with Plotly Dash

<Dashboard Screenshot 1>

- Replace <Dashboard screenshot 1> title with an appropriate title
- SpaceX launch data in real-time.
- Show the screenshot of launch success count for all sites, in a piechart
- Explain the important elements and findings on the screenshot

Show the screenshot of launch success count for all sites, in a piechart

- The flight CCAFS LC-40 is the scored 29.1% as the success launches by site

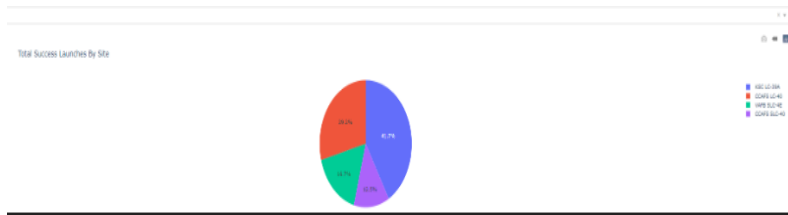


<Dashboard Screenshot 2>

- Replace <Dashboard screenshot 2> title with an appropriate title
- a pie chart visualizing launch success counts.
- Show the screenshot of the piechart for the launch site with highest launch success ratio
- Explain the important elements and findings on the screenshot

Dashboard Screenshot 2

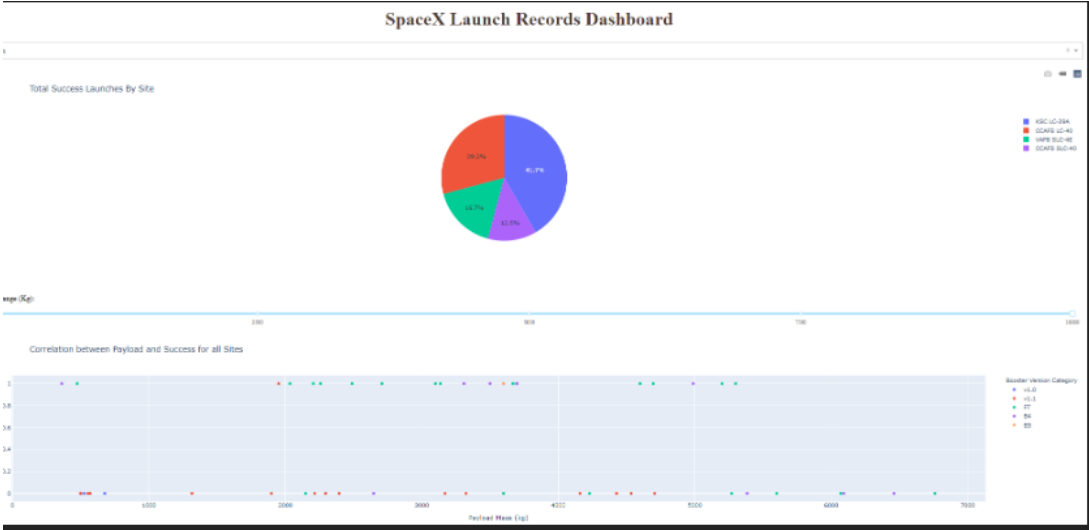
- The flight KSC LC-40 is the scored 29.1% as the success launches by site



<Dashboard Screenshot 3>

- Replace <Dashboard screenshot 3> title with an appropriate title
- **Add a Range Slider to Select Payload**
- Show screenshots of Payload vs. Launch Outcome scatter plot for all sites, with different payload selected in the range slider
- Explain the important elements and findings on the screenshot, such as which payload range or booster version have the largest success rate, etc.

Task 3



Section 5

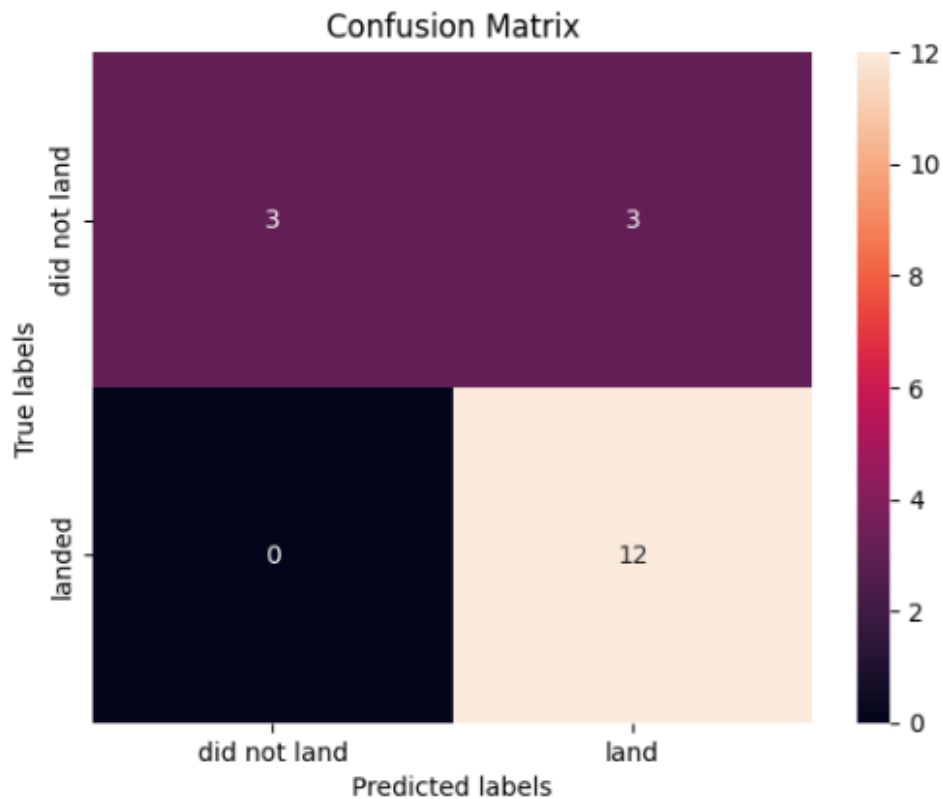
Predictive Analysis (Classification)

Classification Accuracy

- Visualize the built model accuracy for all built classification models, in a bar chart
- Find which model has the highest classification accuracy
- It is **Random Forest Classifier** achieved the **highest classification accuracy**

Confusion Matrix

- Show the confusion matrix of the best performing model with an explanation



- The confusion matrix compares the model's predictions with the actual outcomes.
- True Positives (TP)**: Correctly predicted positive cases
- True Negatives (TN)**: Correctly predicted negative cases
- False Positives (FP)**: Predicted positive but actually negative (Type I error)
- False Negatives (FN)**: Predicted negative but actually positive (Type II error)
- A good model has **high TP & TN** values and **low FP & FN** values.

Conclusions

- Point 1
 - The most frequent landing outcome between **2010-06-04** and **2017-03-20** was **[X outcome]**, showing that SpaceX was consistently achieving this result during the period.
- Point 2
 - Some outcomes such as **[Y outcome]** occurred fewer times, reflecting challenges with early landing attempts.
- Point 3
 - Overall, the results highlight SpaceX's progress — moving from frequent failures to more successful landings as technology and operations improved.

Appendix

- Include any relevant assets like Python code snippets, SQL queries, charts, Notebook outputs, or data sets that you may have created during this project

Thank you!

